# Quiz Answers, Practice Problems Answers

## 1 Quiz Answers

1. What does the following code print?

```
x = 5
y = x + 1
x = 10
print(y)
```

*Answer*

6

2. What is the output of this code?

```
a = 2
b = 3
a = b
b = a
print(a, b)
```

*Answer*

3 3

***Warning***: You will lose points on the midterm if you write:
3, 3
(there is no ',' separating the values displayed by `print`):

3. What is the output of the following code?

```
print("1. ", "3" + "3")
print("2. ", 3 + 3)
```

*Answer*

1. 33
2. 6

4. Circle *all* of the following that are valid Python variable names:

`var2` `var-2` `var+` `var_2` `2var` `var!`

5. The following code produces an error. Explain *why*, and rewrite the last line so it works.

```
course_num = 121
print("HCAI " + course_num)
```

*Answer*

You cannot join a string and an int using +. (It is a type error.)

The fix is to replace the last line with:

`print("HCAI " + str(course_num))`

6. What does the following code print?

```
def g(x):
    y = x + 2
    return y

x = 5
y = g(10)
print("x = ", x)
print("y = ", y + 2)
```

*Answer*

x = 5
y = 14

7. What is the output of this code?

```
def f(x):
    print(x + 5)

y = f(5)
print(y)
```

*Answer*

10
None

8. Circle *all* of the following that are valid Python function names:

`func` `2d_func` `my_func` `_func` `func!` `func-2`

9. Consider the following function:

```
def order(item, size="small"):
    return item + " " + size
```

(a) For each of the following lines of code, put an ✓ next to it if it generates the following output: `coffee medium`

____ `print(order("coffee"))`

✓ `print(order("coffee", size="medium"))`

✓ `print(order("coffee", "medium"))`

(b) For each of the following lines of code, put an ✓ next to it if it generates an error.

✓ `order()`

____ `order("coffee")`

✓ `order(size="medium")`

____ `order(size="medium", item="coffee")`

✓ `order(size="medium", "coffee")`

10. Select what the following code prints, or whether it produces an error. Put an ✓ next your answer.

```
print(not False or True, not (True or False) and (True or not False))
```

____ `True True`

____ `False True`

✓ `True False`

____ `False False`

11. What is the output of the following code? Put an ✓ next your answer.

```
temp = 85
if temp >= 80:
    label = "hot"
if temp >= 60:
    label = "warm"
else:
    label = "cool"
print(label)
```

____ hot

✓ warm

____ cool

12. What is the output of this code?

```
color = "red"
if color == "blue" or "green":
    print("cool color")
else:
    print("warm color")
```

*Answer*
cool color

13. Given the following function, put an ✓ next to all of the print statements that displays: `No input provided`.

```
def greeting(input = None):
    resp = "Hello " + input if input else "No input provided"
    return resp
```

✓ `print(greeting())`

____ `print(greeting("Alice"))`

✓ `print(greeting(""))`

____ `print(greeting(" "))`

____ None of the above: the code produces an error

14. What is the output of the following code?

```
def ticket_price(age, member=False):
    if age < 12:
        price = 5
    elif age >= 65:
        price = 8
    else:
        price = 12
    if member:
        price = price - 2
    return price

print(ticket_price(70, True))
print(ticket_price(30))
```

*Answer*
6
12

15. What is the output of this code?

```
for i in range(3):
    print(i)
```

*Answer*
0
1
2

16. What is the output of the following code?

```
count = 1
n = 10
while n > 1:
    n = n - 2
    count = count + 2
print(n, count)
```

*Answer*
0 11

17. What is the output of the following code?

```
def f(input):
        n = 0
        for x in input:
                if x == "p":
                        n = n + 1
        return n

print(f("philosophy"), f("maryland"))
```

*Answer*
2 0

18. Complete the code in the second box so that it outputs the number of player wins followed by the number of computer wins after the match keeps playing until the player or the computer reaches 5 wins. As a reminder, the `winner` function is provided below.

```
def winner(player, computer):
    if player == computer:
        return "tie"
    elif (player == "rock" and computer == "scissors") or \
         (player == "paper" and computer == "rock") or \
         (player == "scissors" and computer == "paper"):
        return "player"
    else:
        return "computer"
```

```
x = 0
y = 0

while

    computer = random_move()
    # random_move() returns a random valid move:
    # either "rock", "paper", or "scissors"
    result = winner("rock", computer)


        x = x + 1


        y = y + 1

print(x, y)
```

```
x = 0
y = 0

while x < 5 and y < 5:

    computer = random_move()
    # random_move() returns a random valid move:
    # either "rock", "paper", or "scissors"
    result = winner("rock", computer)

     if result == "player":
        x = x + 1

     elif result == "computer":
        y = y + 1

print(x, y)
```

19. What is the output of the following code?

```
print([10, 10] + [1, 2, 3])
```

*Answer*
[10, 10, 1, 2, 3]

20. What is the output of the following code?

```
print([0, 1] * 3)
print(["a"] * 3)
```

*Answer*
```
[0, 1, 0, 1, 0, 1]
['a', 'a', 'a']
```

21. What is the output of the following code?

```
letters = ["x", "y", "z"]
result = letters.append("a")
print(result)
print(letters)
```

*Answer*

None
['x', 'y', 'z', 'a']

22. What is the output of the following code?

```
x = [7, 42]
y = x
x[0] = 10
print("x: ", x)
print("y: ", y)
```

*Answer*

x: [10, 42]
y: [10, 42]

23. What is the output of the following code?

```
arr = []
for i in range(5):
    arr.append(i + 2)
print(arr)
```

*Answer*

[2, 3, 4, 5, 6]

## 2 Additional Practice Problems

1. What is the output of the following code?

```
xs = [1, 2, 3]
ys = xs
ys.append(4)
print(xs)
```

*Output*
```
[1, 2, 3, 4]
```

2. What is the output of the following code?

```
nums = [10, 20, 30, 40, 50]
print(nums[1:4])
print(nums[-2:])
```

*Output*
```
[20, 30, 40]
[40, 50]
```

3. What is the output of the following code?

```
words = ["hi", "hello", "hey", "yo"]
print([w for w in words if len(w) <= 2])
```

*Output*
```
['hi', 'yo']
```

4. What is the output of the following code?

```
grid = [[1, 2], [3, 4]]
total = 0
for row in grid:
    for x in row:
        total = total + x
print(total)
```

*Output*
```
10
```

5. What is the output of the following code?

```
sentence = "the quick brown fox"
words = sentence.split()
print(len(words), words[0], words[-1])
```

*Output*

```
4 the fox
```

6. What is the output of the following code?

```
s = "python"
print(s[0], s[-1], s[1:4], len(s))
```

*Output*

```
p n yth 6
```

7. What is the output of the following code?

```
nums = [3, 1, 2]
print(sorted(nums))
print(nums)
nums.sort()
print(nums)
```

*Output*

```
[1, 2, 3]
[3, 1, 2]
[1, 2, 3]
```

8. What is the output of the following code?

```
def min_max(nums):
    return min(nums), max(nums)

lo, hi = min_max([5, 2, 9, 1])
print(lo, hi)
```

*Output*

```
1 9
```

9. What is the output of the following code?

```
result = []
for i in range(1, 4):
    result.append(i * 10)
print(result)
```

*Output*
```
[10, 20, 30]
```

10. What is the output of the following code?

```
fruits = ["apple", "pear"]
print("apple" in fruits, "grape" in fruits, len(fruits) == 2)
```

*Output*
```
True False True
```

11. What is the output of the following code?

```
xs = [1, 2, 3, 4]
xs.pop(0)
xs.remove(3)
print(xs)
```

*Output*
```
[2, 4]
```

12. What is the output of the following code?

```
print("ab" * 3)
print([1, 2] * 2)
print(3 * "7")
```

*Output*
```
ababab
[1, 2, 1, 2]
777
```

13. What is the output of the following code?

```
print(int("10") + 5)
print(str(10) + "5")
print(int("3") * "ab")
```

*Output*

```
15
105
ababab
```

14. What is the output of the following code?

```
print(8 / 2)
print(len("python"))
print(len([3, 6, 9]))
```

*Output*

```
4.0
6
3
```

15. What is the output of the following code?

```
for value in [0, "", [], None, "hi", [1], 5]:
    if value:
        print("truthy")
    else:
        print("falsy")
```

*Output*

```
falsy
falsy
falsy
falsy
truthy
truthy
truthy
```

16. What is the output of the following code?

```
items = []
if items:
    print("yes")
else:
    print("no")
print(len(items))
```

*Output*

```
no
0
```

17. What is the output of the following code?

```
print(0 or 10)
print("" or "default")
print("hi" or "default")
```

*Output*

```
10
default
hi
```

18. What is the output of the following code?

```
def sign(n):
    if n > 0:
        return "pos"
    elif n < 0:
        return "neg"

print(sign(5))
print(sign(0))
```

*Output*

```
pos
None
```

19. What is the output of the following code?

```
def first(items):
    for x in items:
        return x

print(first([10, 20]))
print(first([]))
```

*Output*

```
10
None
```

20. What is the output of the following code?

```
def winner(player, computer):
    if player == computer:
        return "tie"
    elif (player == "rock" and computer == "scissors") or \
         (player == "paper" and computer == "rock") or \
         (player == "scissors" and computer == "paper"):
        return "player"
    else:
        return "computer"

print(winner("rock", "scissors"))
print(winner("paper", "paper"))
print(winner("scissors", "rock"))
```

*Output*

```
player
tie
computer
```

21. What is the output of the following code?

```
def cycle(round_number, last_computer_move):
    leftover = round_number % 3
    if leftover == 1:
        return "rock"
    elif leftover == 2:
        return "scissors"
    else:
        return "paper"

for r in range(1, 5):
    print(cycle(r, None))
```

*Output*

```
rock
scissors
paper
rock
```

22. This strategy is provided in Project 1. In one or two sentences, explain what it does.

```
def by_round(round_number, last_computer_move):
    if round_number % 2 == 0:
        if random.randint(1, 2) == 1:
            return "rock"
        else:
            return "paper"
    else:
        if random.randint(1, 2) == 1:
            return "paper"
        else:
            return "scissors"
```

*Answer* On even round numbers it returns `"rock"` or `"paper"`, chosen at random, and on odd rounds it returns `"paper"` or `"scissors"`, chosen at random. It ignores the computer's last move.

23. This function is provided in Project 1. Explain what it does and what it returns.

```
def validate_input(player):
    if player == "rock" or player == "paper" or player == "scissors":
        return True
    else:
        print("Invalid input. Please enter rock, paper, or scissors.")
        return False
```

*Answer* It returns `True` when `player` is `"rock"`, `"paper"`, or `"scissors"`. Otherwise it prints an error message and returns `False`.

24. This code will not run. Explain the error and say how to fix it.

```
def greet(name)
    return "Hi " + name
```

*Answer*

`SyntaxError: expected ':'`

The `def` line is missing its colon. Fix: `def greet(name):`.

25. This code will not run. Explain the error and say how to fix it.

```
x = 5
if x > 3
    print("big")
```

*Answer*
`SyntaxError: expected ':'`
The `if` line is missing its colon. Fix: `if x > 3:`.

26. This code will not run. Explain the error and say how to fix it.

```
print("hello)
```

*Answer*
`SyntaxError: unterminated string literal`
The string is missing its closing quote. Fix: `print("hello")`.

27. This code will not run. Explain the error and say how to fix it.

```
x = 5
if x = 5:
    print("five")
```

*Answer*
`SyntaxError: invalid syntax`
`=` assigns a value; a comparison needs `==`. Fix: `if x == 5:`.

28. This code will not run. Explain the error and say how to fix it.

```
def f(x):
return x + 1
```

*Answer*
`IndentationError: expected an indented block`
The body of the function must be indented. Fix: indent `return x + 1`.

29. This code runs, but there is a bug. The expected output is `[1, 2, 3]`, but the actual output is `None`. Explain why there is a bug and how to fix the code.

```
nums = [3, 1, 2]
nums = nums.sort()
print(nums)
```

*Answer* `nums.sort()` sorts in place and returns `None`, so `nums` becomes `None`. Fix: call `nums.sort()` on its own line and then `print(nums)`; or use `nums = sorted(nums)`.

30. This code runs, but there is a bug. The expected output is `6`, but the actual output is `3`. Explain why there is a bug and how to fix the code.

```
def total(numbers):
    for x in numbers:
        s = 0
        s = s + x
    return s

print(total([1, 2, 3]))
```

*Answer* `s = 0` is inside the loop, so it resets every iteration and only the last number survives. Fix: move `s = 0` above the loop.

31. This code runs, but there is a bug. The expected output is `['eggs', 'milk']`, but the actual output is `['eggs']`. Explain why there is a bug and how to fix the code.

```
def add_item(items, x):
    items = items + [x]

shopping = ["eggs"]
add_item(shopping, "milk")
print(shopping)
```

*Answer* `items = items + [x]` makes a new list and points the local name at it; the caller's list is untouched. Fix: use `items.append(x)`, or `return items + [x]` and assign the result.

32. This code runs, but there is a bug. The expected output is `False`, but the actual output is `True`. Explain why there is a bug and how to fix the code.

```
def is_vowel(ch):
    if ch == "a" or "e" or "i" or "o" or "u":
        return True
    return False

print(is_vowel("z"))
```

*Answer* The condition is `(ch == "a") or "e" or ...`, and `"e"` is always truthy, so it is always true. The fix is to replace the if statement with:
`if ch in "aeiou":`
or
`if ch == "a" or ch == "e" or ...`).

33. This code runs, but there is a bug. The expected output is `3`, but the actual output is `1`. Explain why there is a bug and how to fix the code.

```
def count_positives(numbers):
    count = 0
    for x in numbers:
        if x > 0:
            count = count + 1
        return count

print(count_positives([1, -2, 3, 4]))
```

*Answer* `return count` is indented inside the loop, so the function returns after the first number. Fix: move `return count` out of the loop (line it up with `for`).

34. This code runs, but there is a bug. The expected output is the numbers 1 through 5, but the actual output is the numbers 1 through 4 (5 is missing). Explain why there is a bug and how to fix the code.

```
# should print the numbers 1 through 5
for i in range(1, 5):
    print(i)
```

*Answer* `range(1, 5)` stops at 4. Fix: replace `range(1, 5)` with `range(1, 6)`.

35. This code runs, but there is a bug. The expected output is 5, 4, 3, 2, 1 each on its own line, but the actual output is it prints `5` over and over and never stops. Explain why there is a bug and how to fix the code.

```
n = 5
while n > 0:
    print(n)
```

*Answer* `n` never changes, so the loop never ends. Fix: add `n = n - 1` inside the loop.